



BOOK REVIEWS

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The Rise and Fall of Urban Economics: Lessons from San Francisco and Los Angeles, by Michael Storper, Thomas Kemeny, Naji P. Makarem, and Taner Osman. 2015. Stanford, California: Stanford Business Books. 305+xvi. ISBN: 9780804789400. \$60.00.

Cast as a “whodunit,” the authors contrast—over the course of 11 chapters—the economic experiences of the two principal cities in California since about 1970. The authors set out to explain what makes an urban economy prosper (rise) or not (fall). Early on in the book, the authors conclude that the Los Angeles economy has been falling and that the San Francisco economy has been rising. Much of the remainder of the book seeks to explain the divergence and what might be done to better encourage the economy of Los Angeles. The authors attribute the divergence to the different ways in which (1) social and economic networks, (2) the practices of local firms, and (3) the overall “ecology of organizations” were reshaped in these two urban economies. Despite the multiple authors, this book is not an edited collection of papers; the entire manuscript is written as though by a single hand. The authors write well; overall, the book is entertaining and thought provoking. At the same time, as a regional scientist, this book left me wanting more; perhaps this simply reflects the idea that the book, being well-written, leads us to think more clearly about new and important questions. Let me explain.

With its focus on networks, contacts, and organizations, the book is best seen as an application of Economic Sociology: closer to the perspective of Harrison White than say Paul Krugman. Although the authors do discuss the work of some leading scholars in regional science (e.g., Tim Bartik, Gilles Duranton, Ed Glaeser, and Diego Puga), readers expecting to find discussions of markets, prices, investment, transportation costs, and substitution will be disappointed. I imagine that the authors would see roots in Economics; precedents for their work on clustering and specialization would presumably include Alfred Marshall’s treatment of industrial districts. However, industrial districts are not the material that made *Principles of Economics* so important. In this era of “non-compete” clauses and other contracting mechanisms to protect intellectual and other property rights, it is unclear to me how one effectively argues the importance of networks without giving matching consideration to the role of contracts. Coming out of the recession of 2008, there has been some evidence of labor market rigidity. Licensing, contracting, and other impediments to fluidity in the labor market may well be emerging as critical issues. What might the experiences of Los Angeles and San Francisco tell us about this?

To me, the approach of the authors is “process tracing.” More familiar to Political Scientists than Economists, this involves detailing a process (the explanation) that is consistent with all available information and data. At its heart, it is a small-*n* approach: in this case, just two cities. To those of us more used to large-*n* studies—with their samples and statistical inference—the notion of rejecting an argument based on a sole inconsistent fact seems strange. Large *n* or small, the central objective of any book is to be compelling. Presumably, the authors had a target audience in mind: someone not inclined to accept the authors’ approach or conclusion. At the end of this book, the target audience was not yet clear to me. The authors do not apparently seek to engage large-*n* regional scientists; their target group is elsewhere.

Any comparative small-*n* urban study must address several questions. First, why these two particular cities? Second, are these two cities comparable in all important respects other than the differences noted by the authors? Third what is the process (e.g., growth, prosperity, or development) under study here and how and why does it operate as it does? Fourth, what are the critical concepts here and why? Fifth, what would the authors have had to find in order for them to reject the process that they trace out? In my view, the book would have been more compelling if it had a clearer (more systematic) response to these questions.

At first glance, this book would seem to be the stuff of Regional Science. Although the book index does not include an entry for “Regional Science,” the authors have the discipline clearly in mind. On pages 19–21, the authors assert that regional scientists would argue that workers migrate from low-wage regions to high-wage regions and that this migration therefore causes wage differentials to shrink over time. On pages 234–236, the authors return to this same argument. This is a narrow view of what constitutes Regional Science. My fear here is that the authors have, perhaps unintentionally, set up the regional scientist as a straw man: a misrepresentation easier to defeat than an opponent’s real argument. To me, Regional Science today is a big tent. It includes scholars who work in the area of communities, housing, industrial organization, inequality, land use, local environments, neighborhoods, segregation, spatial statistics, sustainability, transportation, and the urban economy. They come from a variety of backgrounds and disciplinary perspectives including economics, geography, planning, political science, and sociology. They apply a wide range of tools including deductive modeling, GIS, qualitative analysis, statistics, and text analysis. To lump all regional scientists together as interested solely in competitive labor market equilibrium does service to no one.

More fundamentally, the strength of Regional Science is the way in which it helps focus the attention of social scientists on the bigger questions of the day and allows the opportunity to think about answers that are not necessarily limited by the perspective of a single discipline. As social science scholars, we should be offering up a clear perspective on how to approach an important intellectual conundrum. This is where the authors appear to flail in my view. Without much prior preparation, they announce that a successful urban economy is one in which the number of high-paying jobs is growing quickly. I don’t fault the authors here. Any number of large city mayors might say exactly the same thing. It makes for good politics. However, does it make for good social science? Scholars dating back to at least Charles Cooley in the 1890s and including Janet Abu-Lughod and Charles Tilly have argued that the imperatives of a global economy largely drive local opportunities for profit. Whether these local opportunities for profit then get spun out in high-wage jobs depends on the ability of local workers to be able to extract part of that profit. So, when the authors assert that San Francisco has created more high-wage jobs, my interest turns to how that happens. In this book, the authors show little interest in how profits are generated and then spent. But perhaps that is to be subject of their next book. If so, I look forward to reading it.

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Sequel to Suburbia: Glimpses of America's Post-Suburban Future, by Nicholas A. Phelps. 2015. Cambridge, Massachusetts: The MIT Press. 231+xii. ISBN: 9780262029834. \$30.00.

This is from the first pages of Nicholas Phelps' *Sequel to Suburbia: Glimpses of America's Post-Suburban Future*: "... it is hard to conceive of the success of postwar American capitalism without these new suburbs. They represent a propitious 'spatial fix' (or geographical embodiment of) for the peculiar strengths of American capitalism during this time and helped generate the profits being produced by American businesses of all sorts during this time. They were the embodiment of America's 'Fordist' mass-production, mass-consumption economy ... The benefits of employment decentralization to the outer suburbs clearly accrued to the federal government in terms of national economic performance" (pp. 1–2).

But what about benefits accruing to the suburbanites who have found a place to live? What about demand? Profits and consumer satisfaction can and usually do go together. "Development interests" (often mentioned) after all are keen to find and please customers (consumers). Perhaps, large numbers of people move to the suburbs because, all things considered, that is their choice (Glaeser and Kahn, 2004). Things happen when there are benefits to buyers as well as sellers. It would be this way even if there was no [United States] "federal government." How do I know? Suburbanization is a widespread phenomenon—even where policies differ significantly from those in the United States (Gordon and Cox, 2011).

The heart of the book is Phelps' three case studies, which stress the various political and administrative failures that stymied hopes for development in a New Urbanist direction. He regrets that actual development was more "horizontal" than "vertical." The author assumes that a New Urbanist path would have been a plausible alternative. In my view, it would have been better had he started by asking essential questions: What do people in cities want? I am fairly sure they want space *and* they want access to their favorite things (including the workplace of one or more household members). They want all this at prices they can afford. How did a resolution come about? Did it involve the auto-oriented development we have? If so, what is the problem?

Phelps worries over the "narrow geographic purview" of most suburbanites and mentions the "legacy of the thinness of the tissue of government within which suburban development has taken place" (p. 71). He cites the "elision of what is being conserved, of environmental and residential amenity with land and property values" (p. 73), one of many ominous "contradictions" that Phelps points to. He prefers arrangements with fewer "parasitic tendencies" (p. 75). A better post-suburban future would be one that develops in light of larger environmental concerns. Throughout, Phelps is content to cite agency or advocacy studies and accepts the view that higher densities would bring about a cleaner environment. But recent academic research is not unequivocal on this point (Glaeser, 2009; Glaeser and Kahn, 2010).

Return to the introductory quotation. Is it useful to speak about a (seemingly inanimate) "federal government" as an important actor in all this? Phelps frequently mentions "the state" and "state intervention." In our federal system, who is that? State governments? Local governments? The courts? The regulatory agencies? All? Some? Which? I prefer a better understanding of the *people in and around politics* with some attention directed at their motivations and limitations. We get the policies we have from the interactions of large numbers of identifiable actors in the political arena.

Consider the Smith–Yandle (2014) "bootleggers and Baptists" view that describes coalitions in which the former have a clear economic interest, whereas the latter have a spiritual interest. When it comes to suburban development, when property owners (including local chambers of commerce) and environmentalists make common cause to make further development costly, both sides have achieved what they seek: property value appreciation and limits on encroachment for the former and environmental preservation for the latter. Phelps notes the "... transitioning from growth-machine style to development-regime style politics" (p. 110). But politics and coalitions are inevitable. We should try to understand how and why they are formed.

"Progrowth" or "antigrowth" city councils do not arrive at their positions based on spiritual convictions. The people who make the rules do so in a political context. Nor is it simply a matter of technocratic study and analysis. The various interest groups are not just bystanders. Development

interests, homeowner interests, and environmental interests can be expected to compete. Then, some find ways to make common cause and coalitions are formed.

Fischel (2015), for example, examines the history of land use policy in the United States and emphasizes the *demand* for various zoning regimes. He considers plausible actors and actions. "The attraction of citywide zoning was the security it gave to the early twentieth-century homebuilders and homeowners" (p. 170). Note that both parties had the same idea. As times changed, the demand explanation continues to hold. Fischel reminds his readers that local politics is inevitably "evolutionary" (p. 277); as there are more local homeowners with greater influence, the transition from "progrowth" to "antigrowth" policy is entirely plausible. The post-1970 years, coinciding with inflation that prompted greater interest in home ownership as a hedge, drove up home prices. This brand of "homevoter" environmentalism is credited with limiting, even reversing, the more "permissive" (progrowth) pre-1970s zoning decisions. Homevoters were able to strike a bootleggers-and-Baptists alliance with organized groups from the environmental movement.

Those who take a NIMBY position have a serious interest in their *immediate* environment. This may not coincide with the "larger environmental impacts of the suburban way of life" (p. 124) that Phelps wishes concerned more people. But most of us are not very good at sorting out our own sensibilities. Can we really untangle the passions from the interests (Hirschman, 1977)? We are prone to ascribing "greater good" motives to our actions. At what point is my interest in neighborhood quality self-serving or a part of my overall interest in the environment? Perhaps, we are all "environmentalists" but not very clear how far the ambit of our concern goes.

Phelps likes to cite "contradictions." Chapter 3 is all about contradictions. These are, of course, an essential element in Marxist dialectics and emphasis on inevitable overproduction, crisis, and revolution. Phelps worries that the suburbanization trends he sees also put us on a road to crisis. His accompanying discussion is (for this reader) not easy to follow. "It is these contradictions of postwar suburbanization that now partly appear to be bound up with a politicization of side effects within a second modernity, not least as a result of the juxtaposition of these contradictions of suburbia with the systematic machinelike properties of the (sub)urban land development imperative, which in large part has continued unabated" (p. 37). There are many others like this; I have a strong preference for authors who do the hard work involved in achieving clarity.

The second half of the book is devoted to three case studies of opportunities missed and what a better post-suburbia might have been. Phelps highlights the stories of three fairly recent American developments: Kendall (Florida), Tysons Corner (Virginia), and Schaumburg (Illinois). The outcomes that the author describes are not the ones he would have preferred. Introducing the first of the case studies, Phelps writes, "Sitting along the I-95-US-1 north-south corridor identified for greater density of development at the state level, the East Kendall suburbs represent one of the very few concerted attempts to fashion a greater density of development and urban amenity. The Kendall downtown experiment has already been hailed as a success of New Urbanism. It is considered part of a broader trend whereby 'an increasing number of Florida cities and counties are confronting the problems of sprawl – traffic jams, lifeless downtown, environmental degradation and sheer ugliness ...'" (p. 87).

For plans and developments closer to Phelps' tastes to succeed, three things have to happen. Planners have to adopt (and enforce) an urban development boundary. They have to see to it that what development there is occurs in higher density "infill" and near transit stations. They also have to hope that enough people will switch to public transit. Phelps writes "... although the post-interstate era has ushered in greater debate on the virtues of, and including, a greater mix of travel modes, there is little momentum for additional investment in mass transit in cities and inner suburbs and less so for the outer suburbs" (p. 67). Phelps evokes the promise of public transit and transit-oriented development. But he appears oblivious to some important facts of life. In 2011, for example, local, state, and federal spending on transit in the United States was 26 percent of the \$207 billion total government effort, while transit's share of worktrips has been stuck at just near 5 percent for the last 20 years. The idea of getting large numbers of Americans back onto public transit, Phelps should know, is an impossible dream. For at least the last 25 years, U.S. public spending has been heavily skewed toward the provision of public transit. Roads and highways have been neglected and transit *capacity* has grown substantially. Nevertheless, auto and transit commuting shares have hardly budged. Most Americans continue to avoid public transit (Winston,

2013). When we look at passenger miles, the contrast is more lopsided: U.S. transit passenger-miles are only 1.7 percent of the total (Cox, 2016). Transit use per capita in the United States has been flat for years (Polzin, 2016).

Should cities be more like suburbs? Should suburbs be more like cities? No one person (or thinker) knows *the* best form of development. Large visions (and visionaries) often let us down; there is nothing like openness to experimentation. Entrepreneurial trial-and-error is the way we discover what works best in which settings. This is not Phelps' view. He expects wisdom to come down from regional and metropolitan leaders who have the uncanny ability to grasp the big picture. Phelps argues for regional government, which will open people's eyes to the bigger problems that they ought to be concerned with. But Progressives' dreams (going back to the turn of the 20th century) of councils of wise technocrats have, in this reviewer's opinion, been upended by the public choice view: officials are with rare exception no better than the rest of us and are (like the rest of us) likely to respond to the incentives and constraints in their lives.

Regional planning and metropolitan government are not new ideas. We see them in many places but we do not see any appreciable positive effect. We live in an exit-voice world. Voice has not worked as hoped. In spite of the election of minority mayors and officials, the poor living in U.S. big cities have not had a good ride. A changed cast at city hall has not made a meaningful difference. Perhaps, finding a way for the disadvantaged to exit is more promising. Moretti (2012) has suggested relocation vouchers for low-income unemployed individuals. School voucher proposals, especially if directed to the poorest families, were proposed in the United States by Milton Friedman over 60 years ago but are still mostly fought and blocked. Most big city office-holders (minority or not) fight to keep the poorest kids in the worst schools. Phelps' prescriptions for greater regional/metropolitan jurisdiction would essentially extend the boundaries of the big-city governments that have hardly reformed anything—and failed our poorest people—into the suburbs.

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The Data Revolution: Big Data, Open Data, Data Infrastructures and Their Consequences, by Rob Kitchin. 2014. Thousand Oaks, California: Sage Publications. 222+xvii. ISBN: 978-1446287484, \$100.

At the outset, I must declare a vested interest in this review. I will soon begin a five-year fellowship in “Transport and Big Data” at the Leeds Institute for Transport Studies (ITS). I will be expected to have decent grasp of the “data revolution,” not only in terms of technology for handling large datasets but also in terms of the wider questions it raises. Like most people, I am learning as I go along and this book greatly helped to place the technical work in its wider context. The data revolution is undeniably driven by technology, meaning the discussion of ethics, for example, often takes a back seat, especially in “data science,” a field that is generally perhaps too macho and technocentric to worry about ethics. What we need therefore is actionable guidance to empower people to navigate the data deluge that is unfolding across all walks of life.

For the impatient reader, *The Data Revolution* does a great job of raising the wider issues associated with the dramatic and still accelerating “datarisation” of life. However, it lacks actionable advice on how to *do* data analysis rather than just read about it. As the book points out, “small data” are not going to disappear and to some extent, Big Data is just an extension of smaller datasets, so one could argue that it is not the book’s role to introduce basic data literacy to the population. I would argue that on the contrary, pointers to newly available open-source resources, at a bare minimum, would help empower people in this area.

An overall synopsis of the book is that it tackles the wider issues related to Big Data, drawing on the social sciences literature. Notable chapters for me included “Conceptualising Data,” which provides a much needed discussion of what data actually are; “Open and Linked Data,” which summarizes and extends Kitchin’s academic work on the subject, and “The Reframing of Science, Social Science and Humanities Research,” which explores the impact of the Data Revolution on academic research.

This review is primarily undertaken in relation to its treatment of Big Data (the subject of chapter 6 but a theme present throughout the book). Why focus on Big Data? The book has already been rightly well reviewed in general terms; I have a strong interest in the subject; and Big Data is probably the fastest moving branch of the revolution with important implications for regional science. It is also an aspect of the data revolution that I have direct experience with, as lead developer of the Propensity to Cycle Tool, an online planning support tool that has been deployed nationally (see the latest version at <http://pct.bike>). This review is therefore inevitably shaped by this context and by the understanding that the data revolution can be harnessed to tackle seemingly intractable issues, such as climate change, for social benefit.

My world-view upon opening this volume was that data are best understood narrowly as a raw material for generating new knowledge in the domain that most interests me: energy use in transport and how to reduce it, as one small part of a grand transition away from fossil fuels in all sectors of the economy (Smil, 2008). *The Data Revolution* has challenged this utilitarian worldview, providing new vocabulary and concepts that will be of real-world use to researchers and practitioners using Big Data, especially when communicating the methods and results to others.

In this context, most people implicitly know that data relate to power (and by extension that Big Data implies great power—just look at Google), but lack the language to articulate the wider economic/political/ethical issues that working with nonstandard datasets entails. Kitchen is frustratingly relativist here. I would argue that people stepping (or more commonly stumbling) into the Data Revolution are better served by pithy and clear guidance rather than an overview of what different people have said in the literature over the years. In the same way that it is the science journalist’s duty to tell us of the best evidence, rather than report all sides of the story (some of which muddy the water), I would argue that it should be the role of the scholar of data to provide normative guidance.

The etymology of Big Data is discussed, running through the usual “three Vs” as well as “exhaustivity” (which Kitchen uses in the sense of covering everything rather than in the sense that Big Data as a byproduct emitted like fumes from a car’s exhaust), resolution, “relationality,” and flexibility. I was disappointed to see little on the fact that Big Data is often a by-product of another process, an important limitation compared with conventional datasets.

Another approach, which is to define large datasets primarily as a technical challenge, was also given only passing reference. From a quantitative geography perspective, Big Data can be defined in terms of the tools that are needed to make sense of them. We acknowledge that in the social sciences, Big Data is used as a catch-all phrase: datasets “that are difficult to analyze using established methods” (Lovelace et al., 2016).

For me the greatest limitation of the book was that it provided little to a nontechnical reader explaining how to actually “do” Big Data. This is a major oversight based on Kitchen’s own insight that open data are only empowering to those who already know how to use it. However, with the parallel revolution of open-source software and the emergence of online communities and software products that make data science accessible to the masses (such as RopenSci and Rstudio), everyday citizens can empower themselves to make sense of the data deluge. I would point people who are interested in processing large datasets toward a list of free data science resources here: <http://datasciencemasters.org/>.

Despite these specific issues surrounding the discussion of Big Data (which should be understood in the context of my worldview, outlined above), the book overall is fantastic. Although Rob Kitchen has written entire papers on many of the topics covered, he does an excellent job avoiding the academic’s curse of going into too much detail at the expense of the Big Picture. And any issues one may have about the coverage of Big Data in the book are more than compensated for by the discussion of data overall. Big Data is after all simply an extension of “small data” (which Kitchen argues cogently should be called “capta” in a fascinating passage), so it is crucial to get to harness these ubiquitous condensed and abstracted chunks of information before trying to tame their comparatively monstrous Big brothers!

The Data Revolution is an elegantly written synthesis of the wider context in which data are becoming to the 21st Century what oil was to the 20th. Too often datasets are seen in purely technical terms. Although technology is crucial for actively participating in the Data Revolution (and I would have liked to have seen more about the open-source communities such as those rapidly coalescing around R and Python), technology without direction is amoral. In summary, *The Data Revolution* can be used as a meta-guidebook for empowering oneself with the conceptual tools needed to understand and use the data flood unfolding worldwide for the greater good.

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Rethinking Global Land Use in an Urban Era, edited by Karen Seto and Anette Reenberg. 2014. Cambridge, Massachusetts: MIT Press. 394+xiv. ISBN: 9780262026901, \$40.00.

The beginning of the 21st century marked a milestone in human history. For the first time, more than half of the world’s population lived in urban areas (3.9 billion). This trend is expected to continue in the foreseeable future with 6.3 billion people living in cities by 2050 (United Nations, 2014). This growth will cause more urban land to be developed during the first 30 years of the 21st century than in all of human history (Angel et al., 2011). Combine this unprecedented urban expansion with global population growth, which is expected to grow from today’s 7.3 to 11.2 billion by 2100 (United Nations, 2015), and we are faced with unprecedented challenges and questions to be asked with respect to land-use in the 21st century. For example, how much living space

will be needed to accommodate this growing population or how much land will be needed to feed such a population? Or how does urban growth in one country impact agricultural production and deforestation in the other parts of the world? To answer these questions, we need to understand the complexity of land competition from social, economic, and environmental perspectives at the local, national, and international levels and the connections between them.

In their edited book, *Rethinking Global Land Use in an Urban Era*, Karen Seto and Anette Reenberg bring together 17 chapters from 50 experts from a variety of fields to explore global land dynamics in the 21st century. The first chapter acts as an introduction and scene-setting to the following chapters: it identifies current trends reshaping land-use locally and globally such as urbanization and the growing integration of economies and markets (e.g., telecoupling, see Seto et al., 2012), but also argues that there is a need to rethink land change science in a time when more and more people are living in cities. Specifically, they argue one should look at land-use through four lenses (which are the major sections in the book): land-use competition; distal land connections; decision making, governance, and institutions; and, finally, urbanization and land-use.

The first section of the book focuses on land-use competition, specifically what types of land-use competition exist (such as forest vs. agriculture or urban vs. agriculture), and discusses how local land-use change is increasingly being caused by global factors (chapter 2). Chapter 3 addresses food security with respect to the growing population and discusses the need for intensification of production. Chapter 4 discusses the issue of finite land resources and competition for land—such as production vs. production (e.g., food vs. fuel) or production vs. conservation (e.g., food production vs. conservation). What is interesting about this chapter is that the competition for land is not just local but also global, due to the growing number of sovereign wealth funds and multinational corporations and the increasing degree of interconnections between places. The section concludes with chapter 5, which offers an in-depth discussion of land-use competition between food production and urban expansion in China, specifically the effects of urbanization on the loss of cultivated land for food production.

The second section of the book explores distal land connections. It opens with a chapter that reviews the globalization of economic flows and the impact of these forces on land-use transitions (i.e., land-use and land-cover change). Chapter 7 introduces applications based on the telecoupling framework to land-change science. It makes a compelling argument for considering not just coupled human–environmental systems (where the focus is on local conditions) but also causes that emanate from distant locations to truly understand land-change. This theme continues in chapter 8, which outlines analytical approaches to study telecoupling, while chapter 9 uses palm oil as a case study of distal land connections. In essence, the consumers of palm oil live far from the source; thus many consumers do not immediately feel the impacts of palm oil production on land-use change.

In the third section of the book, the focus is on decision-making, governance, and institutions. Chapter 10 discusses the emergence of global land governance as a result of land grabbing by foreign investors or governments (see GRAIN, 2008), which is prompting states and global civil society to devise new global land governance instruments, while chapter 11 explores large-scale land (grabbing) transactions with a specific emphasis on the actors and their interactions. Chapter 12 focuses on private market-based regulations (such as the Forest Stewardship Council) and what they mean for land-use governance at the local and international level. The final chapter in this section focuses on changes in land-use governance in an urban era. It discusses how governance mechanisms that manage land-use are changing from territorial organizations to global industries that are tied to specific resource flows between urban and rural areas.

The final section of the book turns to urbanization and land-use. Chapter 14 reviews major contemporary urban patterns and processes related to urbanization, such as central place theory, and shows how advances in technology and infrastructure challenge such established theories. The next chapter discusses how urban land-use is unique in terms of form, size, and shape of cities and asks what will the future hold? Will cities be sprawling or compact? An interesting fact brought up in chapter 15 is that currently less than 5 percent of the earth's surface is urban and with the urban population predicted to grow to 5 billion by 2030, the urban footprint will still be less than 10 percent (Seto et al., 2011). The final chapter in this section proposes a framework that moves away from looking at land as discrete categories but instead as a continuum with respect to sustainable development. The book concludes with a chapter written by the editors, which not only provides a

summary of what was presented, but reemphasizes the interconnected nature of land-use and the need to study future global land change and urbanization from a multidisciplinary perspective.

Overall this is a timely, relevant, and thought-provoking collection of papers which not only explores urbanization and food production using case studies from around the world as well as the connections between cities and distant places, but also lays the foundation for new ways of thinking about land-use sustainability in the coming decades. In my opinion, this book would be a great resource for scholars interested in current state of the art of land-use science and a good textbook for any course exploring land-use and land-cover change in the 21st century.

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Global Entrepreneurship, Institutions and Incentives, by Zoltán J. Ács. 2016. Northampton, Massachusetts: Edward Elgar. 634+xxxiv. ISBN: 9781784718046. \$210.00.

With this edited volume, Zoltán Ács is "closing the books" on the research program "Entrepreneurship, Growth and Public Policy: Prelude to a Knowledge Spillover Theory of Entrepreneurship" that he worked on between 1987 and 2006. The book contains 32 chapters plus an introduction. All chapters except three are co-authored. The total number of co-authors is 35, illustrating his very wide cooperation network. However, some co-authors appear more than once: László Szerb (4 times), Sameeksha Desai, Mark Sanders, Bo Carlsson, David Audretsch, and Pontus Braunerhjelm (3 times), and Utz Weitzel, Lawrence Plummer, Haifeng Qian, Siri Terjesen, and Erkkö Autio (2 times).

It is no easy task to review this volume given its very rich and to a certain degree varied content. The book is organized in six parts and we think that the headings of these six parts give a good overview of the content:

- Part 1 Incentives and the many faces of entrepreneurship*
Part 2 The knowledge spillover theory of entrepreneurship
Part 3 Cities, knowledge and entrepreneurship
Part 4 Countries, institutions and entrepreneurship
Part 5 The entrepreneurial society
Part 6 Institutions, incentives, and public policy

It is an impossible task to review or even mention all the chapters in a book with 32 chapters. Thus, we will limit our comments to seven chapters that we consider to be the most important contributions. We start off with two chapters from part 2 that are co-authored with David Audretsch, Bo Carlsson, and Pontus Braunerhjelm: chapter 6, “The Missing Link: Knowledge Diffusion and Entrepreneurship in Endogenous Growth,” and chapter 7, “The Knowledge Spillover Theory of Entrepreneurship.” (The chapter order is a bit odd, since chapter 7 was published in January 2009 and chapter 6 in February 2010 but of course the publication date says little about when the research was done.) These two chapters illustrate how the black box in new endogenous growth theory can be opened up by providing a microeconomic link between investments in knowledge and economic growth by introducing entrepreneurs as knowledge spillover agents. Aside from the traditional mechanism of innovative entry/product innovation by incumbent firms through investments in research and development (R&D), they introduce a second mechanism of entry by entrepreneurs who do no R&D of their own but use unexploited knowledge generated by incumbents through their R&D and thus function as knowledge spillover agents. It seems obvious that such a mechanism should be included in endogenous growth models, since it is a well-known phenomenon that employees, including innovators, tend to leave the incumbent firm where they work to start a new firm using knowledge they gained as employees. What we have here is nothing less than a contribution to endogenous growth theory.

Another innovative contribution we find in part 6. The title of the chapter is “National Systems of Entrepreneurship: Measurements Issues and Policy Implications” and it is co-authored with Erkkö Autio and Lázló Szerb. A National System of Innovation is defined as a resource allocation system that is driven by opportunity pursuit of individuals through the creation of new ventures, where the outcomes are regulated by country-specific institutional characteristics. Here the authors provide other researchers with an instrument that can be used for comparing the level and direction of entrepreneurship in different countries.

It is well established that the growth and employment effects of entrepreneurship to a large extent is a function of the growth rate of new firms. This has induced policy-makers and researchers in particular to focus on high-growth firms, so-called gazelles. This book contains two chapters analyzing gazelles: “Employment Effects of Business Dynamics: Mice, Gazelles and Elephants,” co-authored with Pamela Mueller in part 3, and the single-authored “High-Impact Firms: Gazelles Revisited” in part 6. In the first paper the authors show that only start-ups with more than 20 and less than 500 employees have persistent employment effects over time and these effects are only found in large diversified metropolitan regions. The second paper illustrates that the lion's share of the employment and revenue growth in the economy comes from high-impact firms. Interestingly, the high-impact firms are seldom really young firms.

An interesting paper in part 4 is “Globalization, Countries, Cities and Multinationals,” which is co-authored with Philip McCann. The focus here is not entrepreneurship in the traditional sense but the relationship between the size of a country, the size of its cities, and the importance of economies of scale in the modern era with globalization and an increasingly important role for multinational firms. In the paper, they reach the interesting tentative conclusion that for industrialized countries the size of a city is nowadays much less important than its level of global connectivity, whereas the size of the city is still dominant in newly industrializing countries. Certainly, this tentative conclusion needs to be verified by future theoretical modeling and formal econometric analyses.

The paper, “What Does ‘Entrepreneurship’ Data *Really* Show?,” in Part 4, which is co-authored with Sameeksha Desai and Leora F. Klepper, is an interesting comparison of two datasets designed to measure entrepreneurship: The Global Entrepreneurship Monitor (GEM) and the World Bank Group Entrepreneurship Survey (WDGES). They find a number of important differences when the two datasets are compared and the magnitude of these differences is related to the local institutional and environmental conditions for entrepreneurs. This is certainly important information for researchers using either of these two datasets in their research.

Zoltán Ács is a highly cited researcher with more than 30,000 citations in *Google Scholar* (mid-April, 2016). The papers in this volume have been cited 3,050 times, although some of his most cited works are excluded. This is equal to more than 95 citations per paper. However, the citation distribution is highly skewed ranging from one paper with 781 citations to three papers with only two citations (the approximate number of citations per year ranges from more than 111

to less than 1). If we instead turn to *Web of Science*, which includes a smaller, more exclusive set of journals, we find there 20 of the chapters in this book. In total they were cited 616 times (early May, 2016). The most highly cited paper is "The Knowledge Spillover Theory of Entrepreneurship." What is interesting here is that Ács receives citations from many fields other than Economics and Management, including areas such as Geography, Planning and Development, Environmental Studies, Urban Studies, Public Administration, and even Psychology. More than 50 percent of the citations do come from Anglo-Saxon countries but there are also a lot of citations from The Netherlands, Germany, Spain, Sweden, and Italy. Overall the bibliometric information indicates a substantial impact across disciplines and countries.

As illustrated above, the recognition of the research documented in this book in the international research community has varied a lot. This, of course, raises questions about why all these different papers were included in the volume. In the preface, Ács writes that, "This volume collects some of my most important publications in the new millennium." This is his personal view, which we respect. However, our quick search with Google Scholar and Web of Science illustrates that the international research community seems to have a somewhat different opinion. There exist a number of publications by Ács that are highly cited but not included in this book.

These examples highlight that this book contains a number of interesting and important papers. However, as a reader one cannot avoid asking oneself why all these papers are included in this volume? Was it really necessary? From the reader's point of view we don't think so. We think that up to nine of the papers could have been excluded. If we instead look upon the book from the editor's point of view we understand why he did choose to include all 32 papers. Ács is "closing the books" for a large and successful research program, which in one sense motivates why all 32 papers are included. Perhaps he also thinks that some of the papers in the book deserve more appreciation among researchers in the field. There is some truth in this. Certainly, the volume serves as a useful handbook of entrepreneurship research.

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Introduction to Regional Economic Development: Major Theories and Basic Analytical Tools, by Mustafa Dinc. 2015. Northampton, Massachusetts: Edward Elgar. 208+xi. ISBN: 9781785361340. \$120.00.

In this tightly written little book, 161 pages of text plus a 22 page methodological annex, Mustafa Dinc sets forth his vision of regional economic development followed by a presentation of analytical tools. Dinc opens the book thus (introduction, p. vii): "Regional and local economic development is a multidimensional and complex process of transforming available resources into desired economic and social outcomes that are sustainable and meet the values and expectations of all participating actors."

Dinc works at the World Bank Statistical Development and Partnership Team. His intention is manifestly not to question or to debate. This is primarily a how-to recipe book with a strong prescriptive bent: regional economic development is a definable process approachable via the right analytical tools. The book's chief strength is the presentation of techniques, models, and methods. However, Dinc has chosen to stay at a fairly general level of analysis and discourse, rarely descending to the messier side of economic development planning.

The book is divided into nine chapters. The first four are broadly conceptual, respectively: (1) *Globalization and regional economies*; (2) *Theoretical evolution of economic development*;

(3) *Governance: institutions, local leadership and the role of regional development agencies*; (4) *Conflict management and regional development*. In each, following a brief review of current thinking, Dinc identifies the ingredients for successful regional economic development. He largely leans toward the endogenous development school of thought with a focus on local actors and capacities. He also cites evolutionary economics (i.e., Ron Martin and other theorists) as a template through which to view the development of regions.

Dinc's approach and language tend to the synoptic, which facilitates generalization, but comes at a price. The reader will find no references to actual regions or case studies. Dinc's "region" remains an abstract object, left undefined, making it difficult to go beyond general prescriptions. He all too often slips into fairly innocuous statements of limited analytical value—two examples: "... effective leadership is a dynamic process driven by interactive local leaders; hence their capabilities should be constantly improved to address the needs of the region" (chapter 3; p. 45); "the role of local (...) leadership is to focus on the current challenges of sustainable development by managing and resolving conflict among stakeholders in the region and promoting creative, technical and institutional solutions" (chapter 4; p. 59).

In chapters 5 and 6, closer to a textbook in style, Dinc introduces familiar models and theories in regional science: chapter 5—*Traditional regional economic development theories* and chapter 6—*Regional agglomeration and industrial clusters*. He presents in turn the great classics of location theory (Von Thünen's rings, Weber's triangle, Hotelling's location model, Christaller's central place theory...), Perroux's growth pole theory, economic base theory, cumulative causation (Myrdal), product life cycles, and the basic principles behind agglomeration economies and industry clusters. Both chapters are on the whole well written, useful references for students seeking a succinct overview of the subject matter.

Chapter 7, *Traditional methods for local and regional economic analysis*, is by far the most substantial, accounting for a quarter of the written text (pp. 96–137). Methods and statistical modeling are Dinc's forte. Dinc presents in turn the basic principles of shift-share analysis, location quotients, economic base analysis, input-output accounts, plus the use of multiple regression techniques as supporting tools for economic base calculations. He visibly wishes to provide the reader with a full toolkit for diagnosing regional economies, using simulated data to demonstrate each method. It is the composite use of all these techniques, Dinc explains, which will allow the analyst to carry out a proper diagnostic of a region's economy and prospects. Shift-share analysis is a particularly useful tool, he argues, for assessing a region's "endogenous" potential via the regional share component. He ends the chapter with a technique with which I am not familiar, Data Envelopment Analysis (DEA), which allows for the integration of multiple data inputs into what he calls DMUs (Decision Making Units), in turn enabling comparisons in the relative efficiency of outcomes using advanced computing techniques. Given my own limitations, I shall not pursue this any further.

Chapter 8, *A regional decision support system*, provides the missing piece for a complete regional toolkit. Dinc presents an Analytic Hierarchy Process (AHP) model which structures the decision process using a multilevel hierarchy plus integrating the information developed in the DEA data framework described in chapter 7. The method asks the decision maker to provide ratio scale comparisons between objectives, and then calculates a vector of weights implied by these comparisons allowing the analyst to prioritize outcomes.

Again, I shall not comment on this method any further as I have never used it. Rather, what interested me was the applicability of all this sophisticated methodological artillery to real-world situations. So far, all demonstrations in the book used simulated data. I was thus pleased to discover a reference to an "actual application" (p. 142) of the full AHP model.¹ I thus diligently did my homework and consulted the article on the Internet. The results of my research, regrettably, did little to cheer me up. First, the reference was incorrect: not in the cited journal as given by Dinc (footnote 1). I eventually located the article in another issue and with a different title²; at least I

¹Dinc, Mustafa, Kingsley E. Haynes, and Murat Tarimcilar. 2003. "A Regional Development Decision Making Model: A Policy Integration Approach," *Annals of Regional Science*, 37(3), 31–53.

²Dinc, Mustafa, Kingsley E. Haynes, and Murat Tarimcilar. 2003. "Integrating Models for Regional Development Decisions: A Policy Perspective," *Annals of Regional Science*, 37(1), 31–53.

believe it to be the same article. This is no minor slipup (in my lexicon), causing me to wonder about other errors.

Coming back to the 2003 article, a rapid reading revealed that chapter 8 is largely a recasting of Dinc's earlier work, which is not necessarily a critique, but leaves me to ask why he did not attribute it. Figures 8.1 and 8.2 are direct transpositions from the 2003 article. We also learn that the secondary data used (before introducing the AHP) are for the State of Virginia and that the AHP application is a simulation, hypothetical weights for socio-political factors assigned by the authors.

In closing, I do not wish to play the bad guy and harp on the book's failings. The book has numerous qualities. I simply wish that the author had chosen to focus on what he does best—methods and statistical modeling—and to include a discussion of data limits and requirements plus a few real-world examples. Even the best technique is ultimately of little use if the data requirements are beyond the reach of most policy analysts. One of the major challenges of regional economic development as a policy science is what to do under conditions of imperfect data and information.

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